WORKED SOLUTION

STEP II Specimen paper, Question 13

Mechanics • Equilibrium and Moments

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q13

The key idea

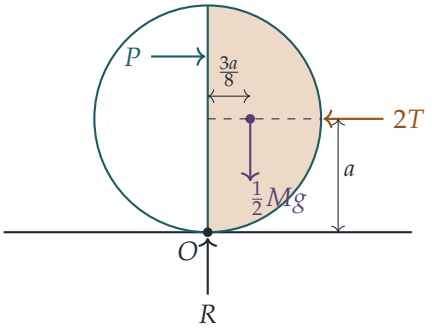
The orange would fall apart by its pieces toppling outwards, each one pivoting about the single point where the sphere touches the table. So we cut the orange in two along one of its slicing planes and take moments for one half about a horizontal line through that lowest point. The weight of the half tries to tip it over; only the ribbon, and nothing else, can hold it back. The forces from the other half can only make matters worse, and that is where the "at least" comes from.

Solution

The model

We treat the chocolate as a rigid uniform solid, the ribbon as light with the same tension T all the way round, and we take the segments to be equal and even in number (a real chocolate orange has twenty), so that one slicing plane through the axis splits the orange into two halves, each a hemisphere-shaped wedge of angle π. At the end we see that the argument works for any group of segments if the cut faces are smooth.

The sphere touches the table at just one point O, the bottom of the axis. Set up axes with O as origin, z vertically up, and x horizontal, pointing from the axis into the half we study (the right-hand half in the figure). The dividing plane is $x = 0$. Let ℓ be the horizontal line through O lying in that plane (the y-axis); we take moments about ℓ, counting a moment as positive if it tends to tip the half outwards, away from the axis.



The forces on one half

The right-hand half has mass $\frac{1}{2}M$, and the external forces on it are as follows.

- **Weight** $\frac{1}{2}Mg$, acting at the centre of mass. Putting $2\theta = \pi$ in the given result, this is at distance
$$\frac{3\pi a \sin(\pi/2)}{16(\pi/2)} = \frac{3a}{8}$$
from the axis, on the side $x > 0$. Its moment about ℓ is $\frac{1}{2}Mg \cdot \frac{3a}{8} = \frac{3}{16}Mga$, in the tipping sense.
- **The table**: a normal reaction and possibly friction, all acting at O. Since O lies on ℓ, these have no moment about ℓ.
- **The ribbon** round its half of the equator. A narrow ribbon presses on the surface normally, with force T/a per unit length (it is like a string over a smooth curve). On the arc between angles $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ the x-components add up to
$$\int_{-\pi/2}^{\pi/2} \frac{T}{a} \cos\phi \; a \, \mathrm{d}\phi = 2T$$
towards the axis, and the y-components cancel by symmetry. (Equivalently, cut the ribbon where it crosses the dividing plane: each cut end pulls with tension T in the $-x$ direction, giving $2T$ again.) All of this force is horizontal and acts at height a, so its moment about ℓ is $2T \cdot a$ in the *restoring* sense.
- **The other half**, acting across the flat face $x = 0$. This is the only awkward force, and we don't need to know how it is distributed.

? **Ask yourself** Why can the force from the other half only help the half to topple?

Split the force on each small patch of the face into a part perpendicular to the face and a part along it (friction, if the cut faces are rough).

- The perpendicular part is in the x-direction. The two halves are only pressed together, never glued, so this part is a push, in the $+x$ direction. It acts at some height $z \geq 0$, so its moment about ℓ is z times its size, which is never negative: it tends to tip the half outwards.

- Any part along the face lies in the plane $x = 0$, which also contains ℓ. A force whose line of action lies in the same plane as ℓ either meets ℓ or is parallel to it, so it has no moment about ℓ.

So the other half contributes a total moment $P_{\text{mom}} \geq 0$ in the tipping sense, whatever the details of the contact.

Moments about ℓ

The half is in equilibrium, so the total moment about ℓ is zero:

$$2Ta = \tfrac{3}{16}Mga + P_{\text{mom}}.$$

Since $P_{\text{mom}} \geq 0$,

$$2Ta \;\geq\; \tfrac{3}{16}Mga, \qquad \text{so} \qquad T \;\geq\; \tfrac{3}{32}Mg,$$

as required. Equality holds exactly when the two halves press on each other with no force at all, so they only just stay together.

Without an even number of segments

If instead the cut faces are smooth, the same calculation works for any wedge of angle 2θ made of whole segments. Take x along its plane of symmetry and ℓ the horizontal line through O perpendicular to that plane. The wedge has weight $\frac{\theta}{\pi}Mg$ at distance $\frac{3\pi a \sin\theta}{16\theta}$, giving a tipping moment of $\frac{3}{16}Mga\sin\theta$. The ribbon on its arc pulls inwards with $2T\sin\theta$ at height a. Smooth faces give only horizontal pushes perpendicular to the faces, and these have a non-negative component in the $+x$ direction, so they again only add to the tipping moment. So $2Ta\sin\theta \geq \frac{3}{16}Mga\sin\theta$, and the $\sin\theta$ cancels to give $T \geq \frac{3}{32}Mg$ once more. Whichever group of segments we choose, the bound is the same.

Checks

- The given centre of mass result at $2\theta = \pi$ gives $\frac{3a}{8}$, the familiar centre of mass of a solid hemisphere measured from its flat face. A Monte Carlo estimate gives $0.376a$ for the half and agrees with the formula for other wedge angles too.
- Summing the ribbon's normal forces numerically over an arc of angle 2θ gives $2T\sin\theta$, for example $2T$ for the half.
- The bound $\frac{3}{32}Mg \approx 0.094Mg$ comes out the same for every wedge angle checked ($2\theta = 0.6,\ \pi,\ 2.4,\ 5$). A bound that depends only on the whole orange, and not on how we choose to cut it, is reassuring.
- Units: the moments are force times length, and a cancels, leaving a force.

Common slips

- Taking moments about the centre of the sphere, or about a point on the equator. Then the unknown force from the table, or from the other half, enters the equation.

The point O is chosen precisely because the table's force has no moment about a line through it.

- Saying the ribbon's force on the half is T. The ribbon pulls on both ends of the half's arc, so the resultant is $2T$.
- Setting the force between the halves equal to zero without justification. That is what gives equality. The "at least" comes from noticing that this force can only push.

Beyond STEP

This is a statics problem with an unknown internal force, handled by choosing an axis that removes the unknowns we can't find and then using an inequality (contact forces push and never pull) to deal with the rest. The same reasoning lies behind the analysis of masonry arches and domes, where a structure stands up only if a line of thrust can be found that stays inside the material. The centre of mass result itself is a standard triple integral in spherical polar coordinates, well worth deriving. For more on rigid-body equilibrium, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

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